2020 Physics Cup Solutions

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June 11, 2026

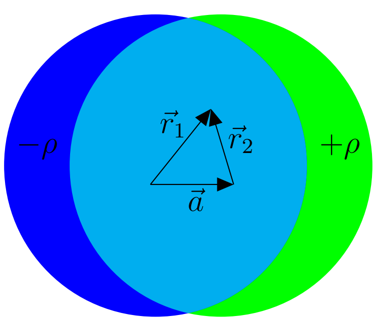
Problem 1

Two metal balls, both of mass M and radius R, are connected with a rigid dielectric weightless rod of length L, assume that $L \gg R$. The system is placed into a uniform external electric field of strength E; neglect gravity and friction forces. Find the period of small oscillations of this system around its stable equilibrium state.

Solution

Claim 1. The spheres form electric dipoles of moment $\vec{d_e} = 4\pi\epsilon_0 R^3 \vec{E}$ under a uniform field $\vec{E}$.

Proof. Positive charges started accumulating on the upper half of the sphere while negative charges start accumulating on the lower half of the sphere as the electric field redistributes the charges. We can model these charges by using the principle of superposition, i.e. Two charged balls of radius R and homogeneous volume charge densities $+\rho$ and $-\rho$ that are separated by a small distance a.



Writing Gauss's law for this spherical surface yields $4\pi R^2 \epsilon_0 E = \frac{4}{3}\pi R^3 \rho$, from where

$$\vec{E} = \frac{\rho \vec{r}}{3\epsilon_0}.$$

Writing two 2D vectors on the $x-y$ plane $\vec{r}_1$ and $\vec{r}_2$ at a certain point with the vector $\vec{a}$ allows us to write

$$\vec{E} = \frac{\rho}{3\epsilon_0}(\vec{r}_1 - \vec{r}_2) = \frac{\rho}{3\epsilon_0}\vec{a}.$$

This is a constant vector, which in the limit of $a \to 0$ and $\rho \to \infty$ tell us that

$$\frac{\rho}{3\epsilon_0} a = E \implies \rho a = 3E\epsilon_0.$$

The sphere will accumulate a surface charge density $\sigma = \rho d$ where $d = a\cos\theta$ where θ is a polar angle of deviation from the electric field. By uniqueness theorem, this is the only possible charge distribution. As $\vec{d_e} = \sum q_i \vec{r_i}$, then $\vec{d_e} = 4\pi\epsilon_0 R^3 \vec{E}$. $\square$

The electric field from a single dipole well known to be given as

$$\vec{E} = \frac{1}{L^3}(3(\vec{d_e} \cdot \vec{e_r})\vec{e_r} - \vec{d_e}).$$

The total energy from the interaction between two dipoles can then be written as (as there are two induced dipoles)

$$\Pi = -2\vec{d} \cdot \vec{E} = \frac{2}{L^3}(\vec{d_e} \cdot \vec{d_e} - 3(\vec{d_e} \cdot \vec{e_r})(\vec{d_e} \cdot \vec{e_r})).$$

If the dumbbell is oriented at an angle φ from its equilibrium position, then the dumbbell will make an angle of φ along their radial vector direction $\vec{e_r}$. Thus,

$$\Pi(\varphi) = \frac{8\pi_0 R^6 E^2}{L^3}(1 - 3\sin^2\varphi) \approx \frac{8\pi_0 R^6 E^2}{L^3}(1 - 3\varphi^2).$$

By Kalda Mechanics method 6, if the potential energy is described by one generalized coordinate, then the equation of motion $\ddot{\varphi} = -\Pi'(\varphi)/\mathcal{M}$ where $\mathcal{M}$ is the total moment of inertia of this system or $\mathcal{M} = 2 \cdot 2M\left(\frac{L}{2}\right)^2 = ML^2$. Thus, our equation of motion is

$$\ddot{\varphi} = -\frac{\Pi'(\varphi)}{\mathcal{M}} = -\frac{48\pi_0 R^6 E^2}{ML^5} \implies T = 2\pi\sqrt{\frac{ML^5}{48\pi_0 R^6 E^2}} = \boxed{\sqrt{\frac{\pi ML^5}{12\pi_0 R^6 E^2}}}.$$

Remark. I think this problem, overall, is quite simple if you knew about the fact that charges will redistribute on the sphere under the presence of an electric field. The main thing that is quite tricky, however, is that it is quite easy to make a mistake and get an incorrect numerical factor to your answer. In fact, it took me three tries to actually get the answer correctly. The official solution does this problem by analyzing torques on the spheres, but I think analyzing potential energy is much slicker.

Problem 2

Solution

From the given profile $y(x) = -(x/\lambda)^{3/2} h$, the n-th step (where $y(x_n) = -nh$) is located at $x_n = \lambda n^{2/3}$, so the inter-step distance is

$$d_n = x_{n+1} - x_n \approx \frac{2}{3}\lambda\, n^{-1/3} \equiv a\, n^{-1/3}, \qquad a = \frac{2}{3}\lambda = 30\ \mu\text{m},$$

in agreement with the EuPhO result. The key property of this law is

$$d_n^{-3} = \frac{n}{a^3} \quad \Longrightarrow \quad d_{n+1}^{-3} - d_n^{-3} = \frac{1}{a^3} \quad \text{for every } n,$$

including $n = 0$: the first step borders the facet, corresponding to $d_0^{-1} = 0$.
As per the solution to the EuPhO problem, the energy change (per unit length of the step) due to a small outward displacement δ of step n is

$$\epsilon_n(\delta) = \mu\left((d_n + \delta)^{-2} - d_n^{-2} + (d_{n+1} - \delta)^{-2} - d_{n+1}^{-2}\right) \approx 2\mu\left(d_{n+1}^{-3} - d_n^{-3}\right)\delta = \frac{2\mu}{a^3}\,\delta.$$

Such a displacement deposits the volume $h\delta$ per unit length onto the surface, so the cost of adding material is the same everywhere on the staircase: the chemical potential per unit volume is

$$\mu_V = \frac{2\mu}{a^3 h}.$$

This uniformity is precisely the equilibrium condition for the curved part of the surface. Its value, however, is fixed by the equilibrium of the facet, which is where β enters.
Suppose we remove the topmost atomic layer of the crystal. In cross-section this layer is a strip of width equal to the facet width W (up to corrections of order $\lambda \ll W$), bounded by one step on each side, so removing it releases, per unit length in z,

$$\Delta U_{\text{steps}} = -2\beta - \frac{2\mu}{d_1^2} \approx -2\beta.$$

Since the crystal is macroscopic, removing one layer leaves the staircase shape (and λ) unchanged; the removed volume hW is simply redeposited on the staircase by shifting the steps outward by amounts δ_n with $\sum_n \delta_n = W$, at the cost

$$\Delta U_{\text{redeposit}} = \frac{2\mu}{a^3}\sum_n \delta_n = \frac{2\mu W}{a^3},$$

independently of how the material is distributed among the steps. In equilibrium this operation must be energy-neutral:

$$2\beta = \frac{2\mu W}{a^3} \quad \Longrightarrow \quad \mu = \frac{\beta a^3}{W} = \frac{8\beta\lambda^3}{27W}.$$

It remains to read W off the graph. The right edge of the facet is at $x = 0$, and the profile reaches $-1\ \mu$m exactly at both corners of the plot: on the right, $y(10\text{ mm}) = -(10^4/45)^{3/2} \times$

$0.3\ \text{nm} \approx -1.0\ \mu\text{m}$. The left branch reaches $-1\ \mu\text{m}$ at $x = -20$ mm, so with the same λ its facet edge lies at $x = -10$ mm, and the facet occupies $-10\ \text{mm} < x < 0$:

$$W = 10\ \text{mm}.$$

(The flat region appears wider only because the $x^{3/2}$ droop is invisible on the scale of the graph for the first few millimetres past each edge.)
Numerically,

$$\mu = \frac{\beta a^3}{W} = \frac{(4 \times 10^{-15}\ \text{J/m})\,(3 \times 10^{-5}\ \text{m})^3}{10^{-2}\ \text{m}} \approx 1.1 \times 10^{-26}\ \text{J} \cdot \text{m}.$$

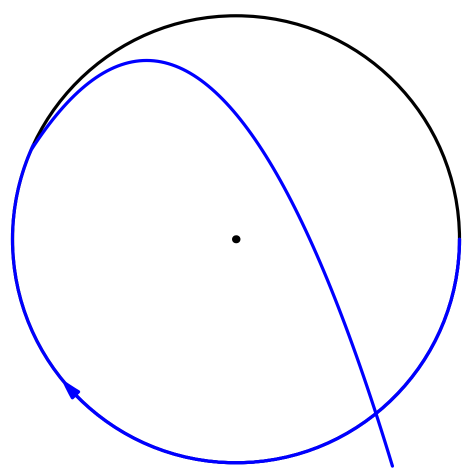
The neglected interaction term is justified *a posteriori*: $\mu/d_1^2 \approx 1.2 \times 10^{-17}\ \text{J/m} \ll \beta$.

Problem 3

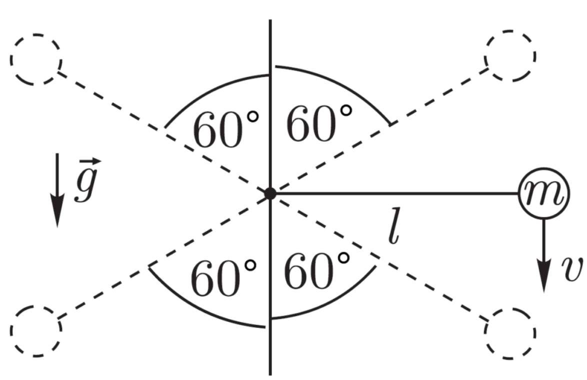
A small ball of mass m is suspended on an in-extensible massless thread of length l. The thread is very thin, flexible, and has a breaking force limit of $T_b = 9mg$, where g denotes the free fall acceleration. In the initial moment the thread is horizontal as shown in the figure, and the ball velocity $v = v_0$ is directed downward. It is known that the thread breaks in one of the four positions shown in the figure with dashed lines, when the angle between the thread and the vertical is 60°; friction is to be neglected. Find all the possible values for v_0. (A solution is considered to be correct only if **all the suitable values** are algebraically expressed, and no extra/invalid values are included.)

Solution

Let us first think of the different ways the string can break. One way is that the centripetal acceleration of the small mass is very big causing for the tension to over-exceed $9mg$. The other way is that the initial velocity of the ball is small enough such that it departs from a circular trajectory and then goes into a parabolic trajectory. An example of such a trajectory is shown below:



So, even though this problem may have looked simple at first, it is actually much more intuitive as we have to figure out how to deal with different trajectories. Let us then first consider when the centripetal of the small mass is very big.



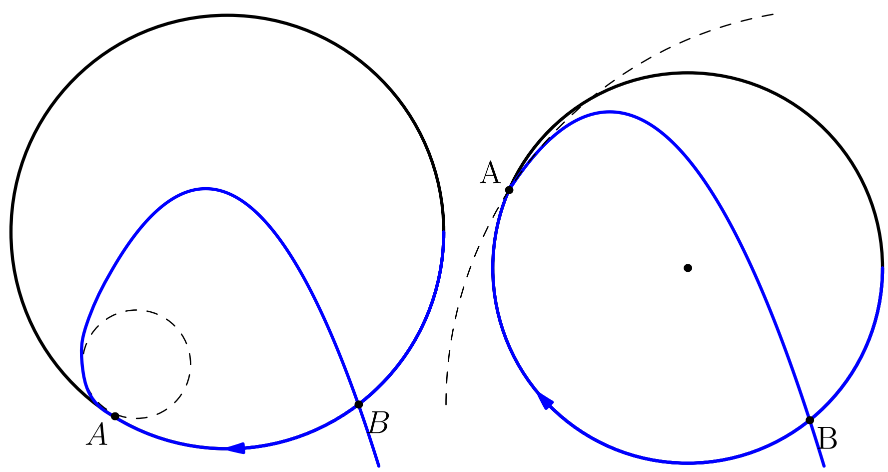
Since the centripetal acceleration is very big, the ball will break at the very first point of breakage (i.e the bottom right point). Let θ where $\theta < 90^\circ$ be the angle from the vertical. By conservation of energy, we can write

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv^2 + mg\ell\cos\theta \implies v_0^2 = v^2 + 2mg\ell\cos\theta.$$

We can balance forces in the radial direction $\hat{r}$ for the case when the tension becomes $9mg$ (because the thread will break then) such that

$$\frac{mv^2}{\ell} + mg\cos\theta = 9mg \implies \frac{m(v_0^2 + 2mg\ell\cos\theta)}{\ell} + mg\cos\theta = 9mg.$$

Plugging in when $\theta = 60^\circ$ gives us $v_1 = \sqrt{15g\ell/2}$. When $v \leq \sqrt{3g\ell}$[1], the tension at the first point will become less than $9mg$, so v_1 is the only case when the velocity is high. We now need to consider when $v \leq \sqrt{3mg}$ (i.e parabolic trajectories). Let the ball depart from it's initial circular at point A, and start moving in a parabolic trajectory until it re-meets the circle at point B.



Before the ball goes into parabolic orbit, the radius of curvature of the path must be smaller than the radius of curvature of the parabola at that point. Otherwise, the centripetal acceleration will be too small to keep the ball still in motion. When the ball departs, the radius of curvature has to be very big so that the tension in the string (because of the centripetal force) can decrease and allow for the ball to then go into parabolic orbit. This is also the same point at where the radius of curvature of the parabola is the same as that of the trajectory.

The ball will only leave at point A when it has accumulated some inward radial velocity (when the radius of curvature is very big and hence there is a very little amount of centripetal force). After some time Δt, the speed of the projectile after it leaves point A will be $|v|_A - \Delta|v|$. This is because the gravity decelerates the mass but the component of acceleration that is perpendicular to the mass will increase (due to a smaller radius of curvature) and therefore pull the mass inwards. This means that the vertex of the parabola will always

[1] This velocity value can be found from manipulating equations with the $F = ma$ equation.

be inside of the circle. Also, note that the circle of curvature will be tangential to the initial path of the parabolic motion of the mass because of this.

Theorem. If a parabola intersects with a circle at four different points, the sum of the polar angles with respect to the axis of symmetry of the circle will be

$$\sum_{i=1}^{4} \varphi_i = 2\pi n, \ \forall n \in \mathbb{Z}.$$

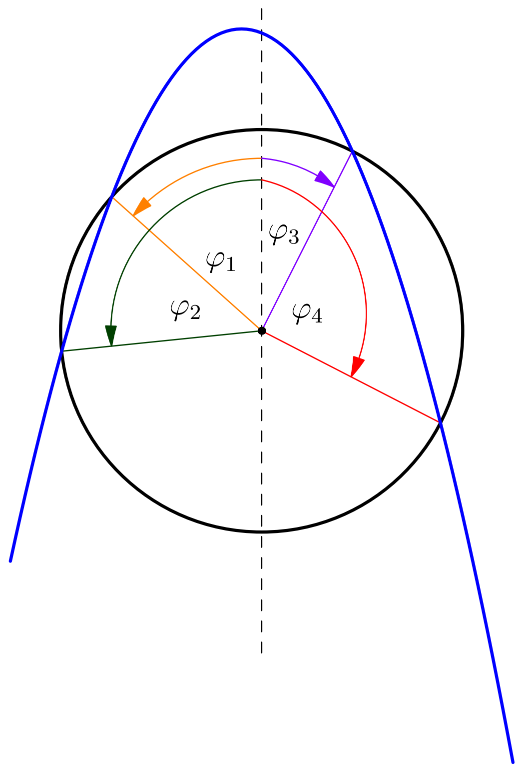
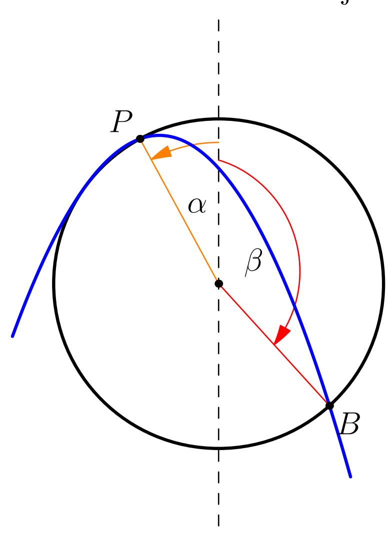


Figure 1: An image of the intersection of a parabola with a circle.

Proof. WLOG, consider a unit circle centered at the origin in the cartesian plane with an equation of $x^2 + y^2 = 1$. Recreating the unit circle in the complex plane gives us the formula of the circle to be $z\bar{z} = 1$ where $z = x + iy$ and $\bar{z} = 1/z$ is its complex conjugate. Transforming coordinates into complex yields, $x = \frac{1}{2}(\bar{z} + z)$ and $y = \frac{i}{2}(\bar{z} - z)$. The parabola equation is $y = ax^2 + bx + c$ so substitution tells us $i(\bar{z} - z) = a(\bar{z} + z)^2 + 2b(\bar{z} + z) + 2c$. The intersection points will satisfy a fourth degree equation in terms of z so there are four roots z_1, z_2, z_3, and z_4. All roots are on the unit circle and are therefore rotations due to Euler's identity $e^{i\varphi_i}$. Vieta's tells us $z_1 z_2 z_3 z_4 = 1$, hence $e^{i(\varphi_1+\varphi_2+\varphi_3+\varphi_4)} = e^{i\left(\sum_{i=1}^{4} e^{i\varphi_i}\right)} = 1$. Rewritten, this is $\sum_{i=1}^{4} \varphi_i = 2\pi n, \ \forall n \in \mathbb{Z}$. $\square$

This theorem is valid only if the parabola intersects with the circle at 3 different points. Therefore, let us imagine moving the parabola while the circle is at rest until the three intersection points merge into one intersection point which we will call P. By our discussion in the second page, the radius of curvature of the trajectory at P will be the same as that of the parabola. This means that we can redraw our trajectory to be as shown below



Now, we say that $3\alpha - \beta = 2\pi n$ where β is the polar angle of the points of intersections or $\alpha = 60°, 120°, -120°$. Upon solving gives us three different angles, $\alpha_1 = 20°, \alpha_2 = 40°, \alpha_3 = 80°$. The velocity of the mass when they leave the circular trajectory can be equated to the centripetal force or $mv^2/\ell = mg\cos\alpha \implies v = \sqrt{g\ell\cos\alpha}$. Once again applying energy conservation (refer to page 1) and substituting the value of v tells us that $v_0 = \sqrt{3g\ell\cos\beta}$.

Hence, the possible values for v_0 are then:

$$\boxed{v_0 \in \{\sqrt{15g\ell/2}, \sqrt{3g\ell\cos(20)}, \sqrt{3g\ell\cos(40)}, \sqrt{3g\ell\cos(80)}\}}$$

Problem 4

The space between two parallel infinite metallic plates is filled with homogeneous material of heat conductivity κ. The heat conductivity of the metallic plates is much larger than κ, the distance between the plates is L. One of the plates is kept at temperature T_1, and the other – at temperature T_2. There is a spherical cavity of radius R in the medium between the plates, at equal distance $L/2$ from the both plates; assume that $R \ll L$. Initially, the cavity is filled with a gas of negligibly small heat conductivity (much smaller than κ). By how much will change the total heat flux P between the plates (measured in watts) if the spherical cavity is filled with a material of very large heat conductivity (much larger than κ)?

Solution

As most people are more familiar with electrostatics, we can move from thermology to electrostatics, derive the answer there, and translate back. Note that temperature T plays the role of the electrostatic potential φ: comparing Fourier's law $\vec{J} = -\kappa\nabla T$ with $\vec{D} = -\varepsilon\nabla\varphi$, the heat flux density $\vec{J}$ corresponds to the displacement field $\vec{D}$, and the conductivity $\kappa(\vec{r})$ corresponds to the permittivity $\varepsilon(\vec{r})$. Consequently, total heat flux corresponds to charge:

$$q = \oint \vec{D} \cdot d\vec{A} \quad \longleftrightarrow \quad P = \oint \vec{J} \cdot d\vec{A}.$$

The equivalent electrostatic configuration is therefore a dielectric ball between two plates held at fixed potentials φ_1 and φ_2. In the uniform applied field, the ball acquires a dipole moment, just as in Problem 1.

A sphere of conductivity κ_s embedded in the medium with unperturbed gradient $G = |T_1 - T_2|/L$ perturbs the temperature outside itself as a dipole:

$$T = -Gr\cos\theta + \beta R^3 G \frac{\cos\theta}{r^2}, \qquad \beta = \frac{\kappa_s - \kappa}{\kappa_s + 2\kappa},$$

matching the polarizability of a dielectric ball of permittivity ε_2 in a medium ε_1. The corresponding thermal dipole moment is $p = 4\pi\kappa\,\beta R^3 G$. Since $R \ll L$, the sphere sits in an essentially uniform gradient, and image corrections from the plates modify p only at higher order in R/L. For the gas-filled cavity ($\kappa_s \to 0$) we get $\beta = -1/2$, while for the filling of very large conductivity ($\kappa_s \to \infty$) we get $\beta = 1$.

To find how the dipole changes the total flux, note first that a free dipole's field carries zero net flux through an infinite plane: the forward flux through the region of the plane nearest the dipole is exactly cancelled by the return flux farther out. The effect therefore comes entirely from the boundary conditions at the isothermal plates. We use Green's reciprocity in the electrostatic picture: a point charge q at height z between two grounded plates induces a total charge $-qz/L$ on the upper plate (apply reciprocity with the auxiliary uniform-field potential $\varphi = Vz/L$). Hence a perpendicular dipole $p = q\,\delta z$ induces an additional charge p/L flowing onto one plate and off the other. Translating charge back into heat flux, the sphere changes the total flux between the plates by

$$\delta P = \frac{p}{L} = 4\pi\kappa\,\beta R^3 \frac{|T_1 - T_2|}{L^2}.$$

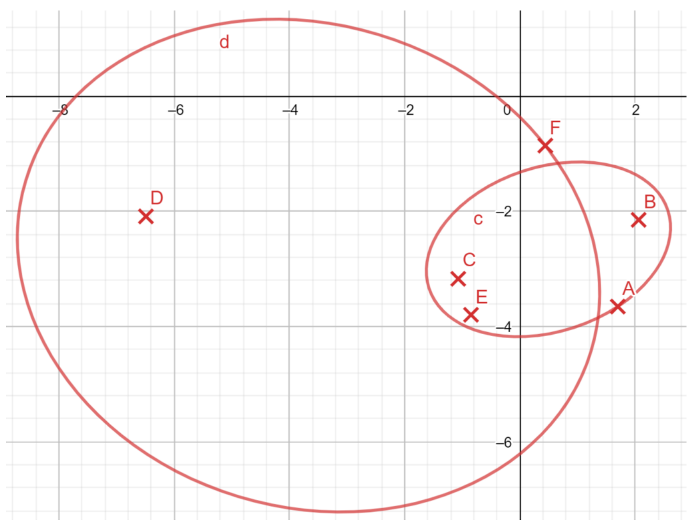
The total flux is thus $P = P_0 + \delta P$, where P_0 is the flux of the unperturbed slab; only δP depends on the filling of the cavity. Replacing the cavity ($\beta = -1/2$) by the highly conducting material ($\beta = 1$) therefore changes the flux by

$$\Delta P = 4\pi\kappa R^3 \frac{|T_1 - T_2|}{L^2}\left(1 + \frac{1}{2}\right) = \boxed{6\pi\kappa R^3\,\frac{|T_1 - T_2|}{L^2}}.$$

As a sanity check, the empty cavity reduces the flux ($\beta < 0$) while the highly conducting sphere increases it ($\beta > 0$), as expected.

Problem 5

The two ellipses depicted in the figure below, available as a Geogebra file, represent real images of circles, created by an ideal thin lens. Both the ellipses and the main optical axis of the lens lay in the plane of the figure. Reconstruct the position of the lens (i.e. the position of the centre and the orientation). Remark: it is recommended to use Geogebra, https://www.geogebra.org/, either as an online service, or as a downloaded application. In you construction steps, you may use everything what Geogebra has to offer: drawing a line through two given points, drawing parallel and perpendicular lines to a given line through a given point, drawing tangents through a given point to a given shape (i.e. a given curve), finding intersection points of shapes (i.e. of curves or lines) , drawing circles with a given centre through a given point, or through three given points, or through two diametrically opposite points on it, drawing ellipse with given foci through a point laying on it, or through five given points. Trial-and-error approach (for instance, displacing a certain shape until three certain intersection points lay on a straight line) is not accepted. You are welcome to send me the `.ggb` files together with the solution. Alternatively, you may print the ellipses onto a sheet of paper and make constructions by hand using a ruler and a compass (be aware that there may be no solution if you start with arbitrarily positioned two ellipses of arbitrary shape, so make sure to either print the drawing, or take the initial shapes close enough to the provided one).

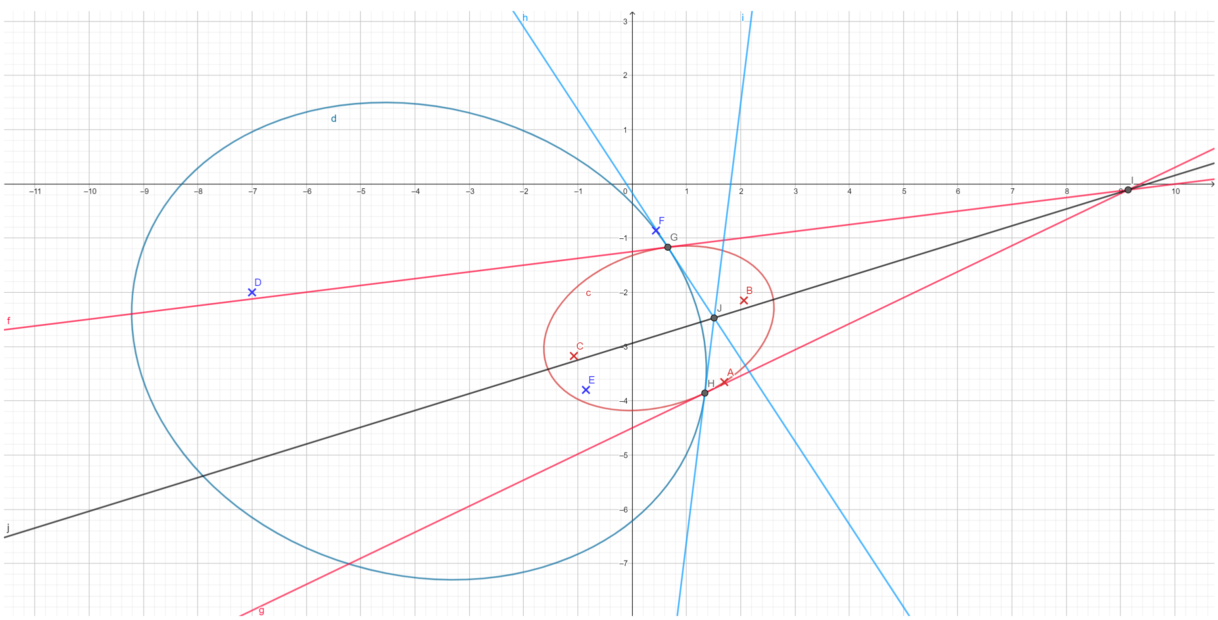


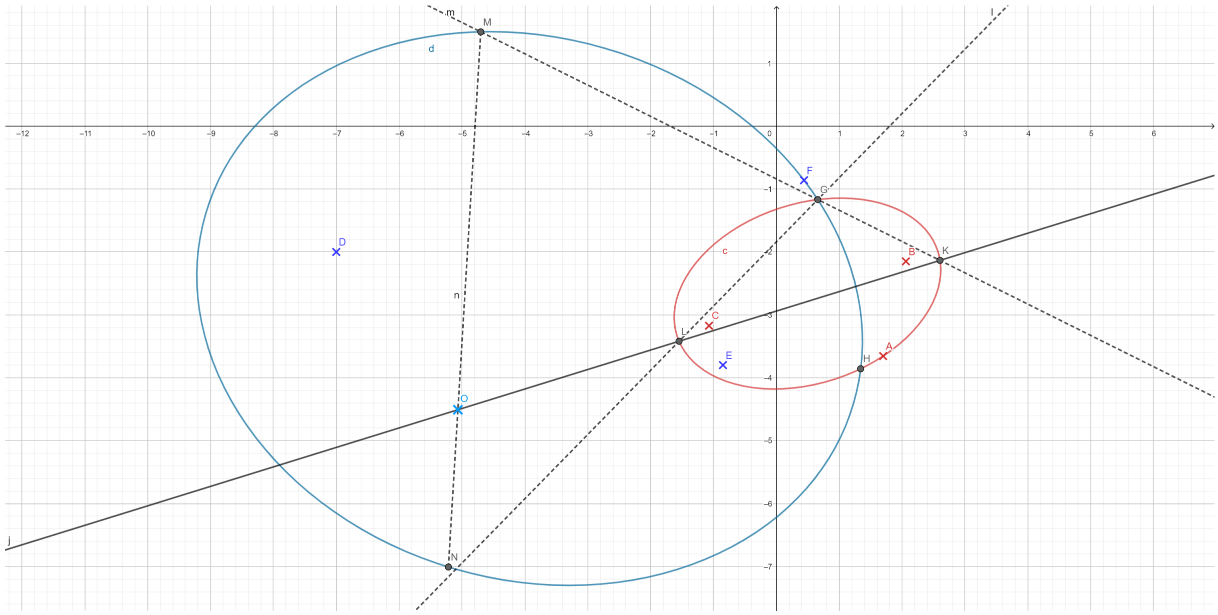
Solution

Let us break this construction down to smaller steps. https://youtu.be/UA8CYEeaog8.

Finding the Center of the Original Circle

For simplicity, let us make the bigger ellipse (d) into a blue color and the smaller ellipse (c) into the same red color. Define points G and H to be the intersection between the blue and red ellipses as shown below. If Γ_1 and Γ_2 pertains to the original circles and Γ_1' and Γ_2' pertains to the transformed ellipses, then the center of Γ_1 can be found on the line ℓ that is created by making tangents with respect to the surfaces of Γ_1' and Γ_2' and connecting their intersections. Now let K and L be the endpoints on the semi-major axis of Γ_2'. We can extend line $\overline{KB}$ until it intersects with Γ_1' at point M. Similarly, we can extend line $\overline{GL}$ until it intersects on Γ_1' at point N. Note that in the original circle, N and M will be diametrically opposite points. So by drawing MN, we then can find the original center of the circle to be at point O or the intersection of MN and KL.




m
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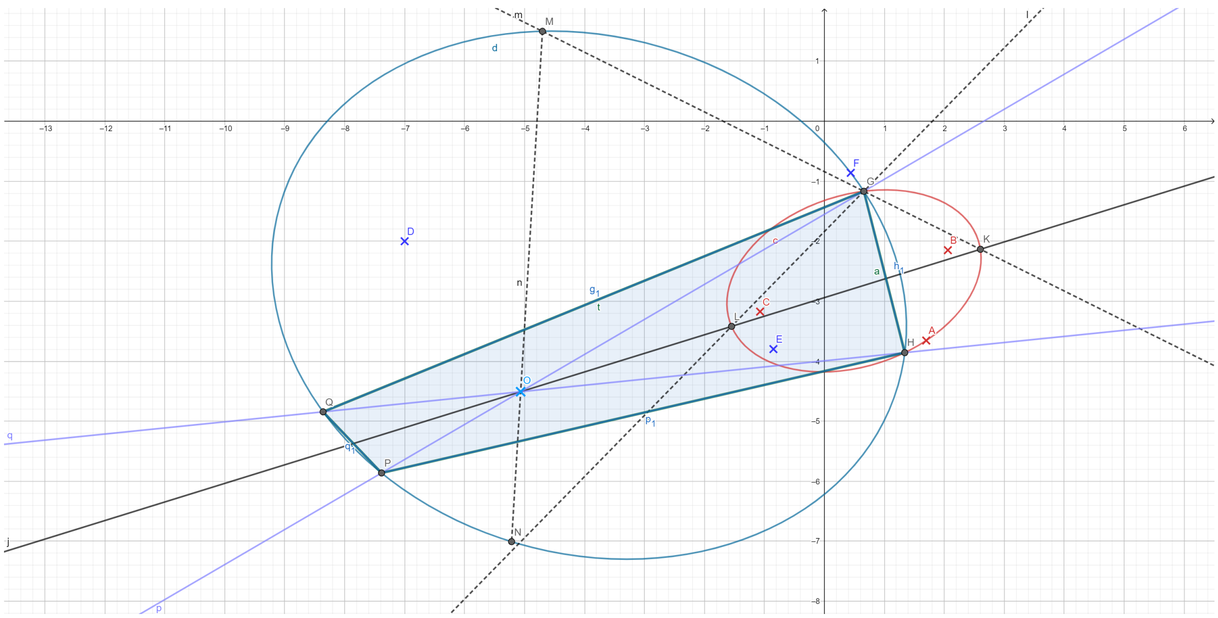
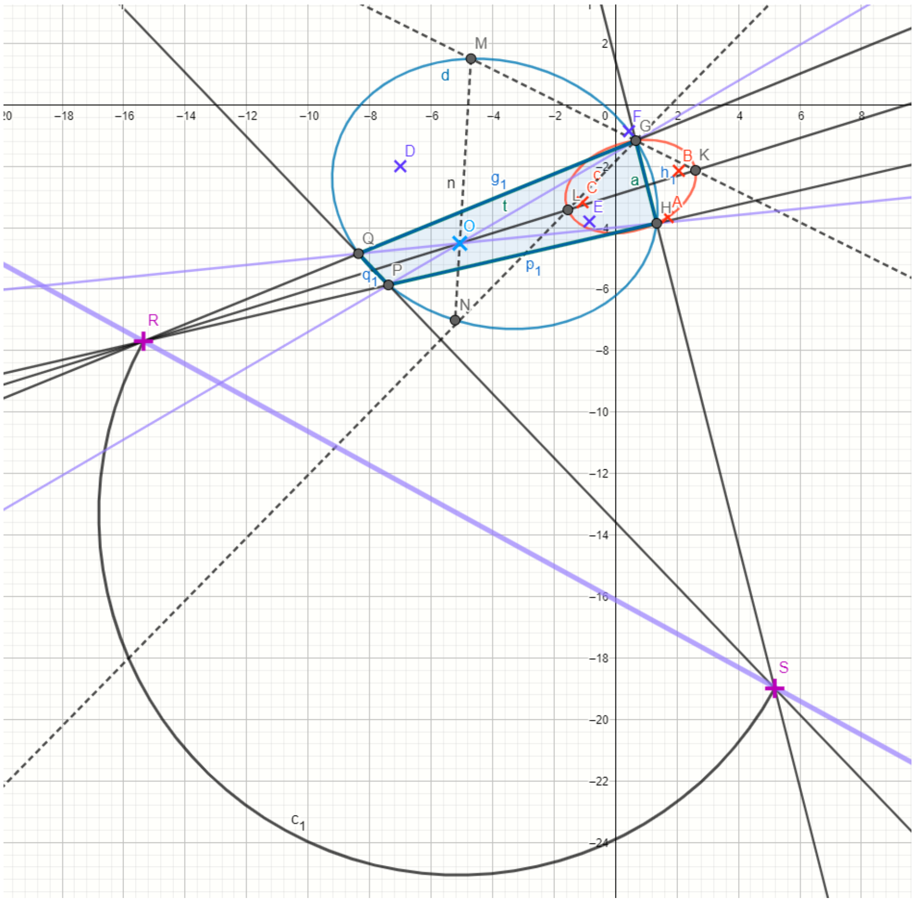
Constructing the Focal Plane

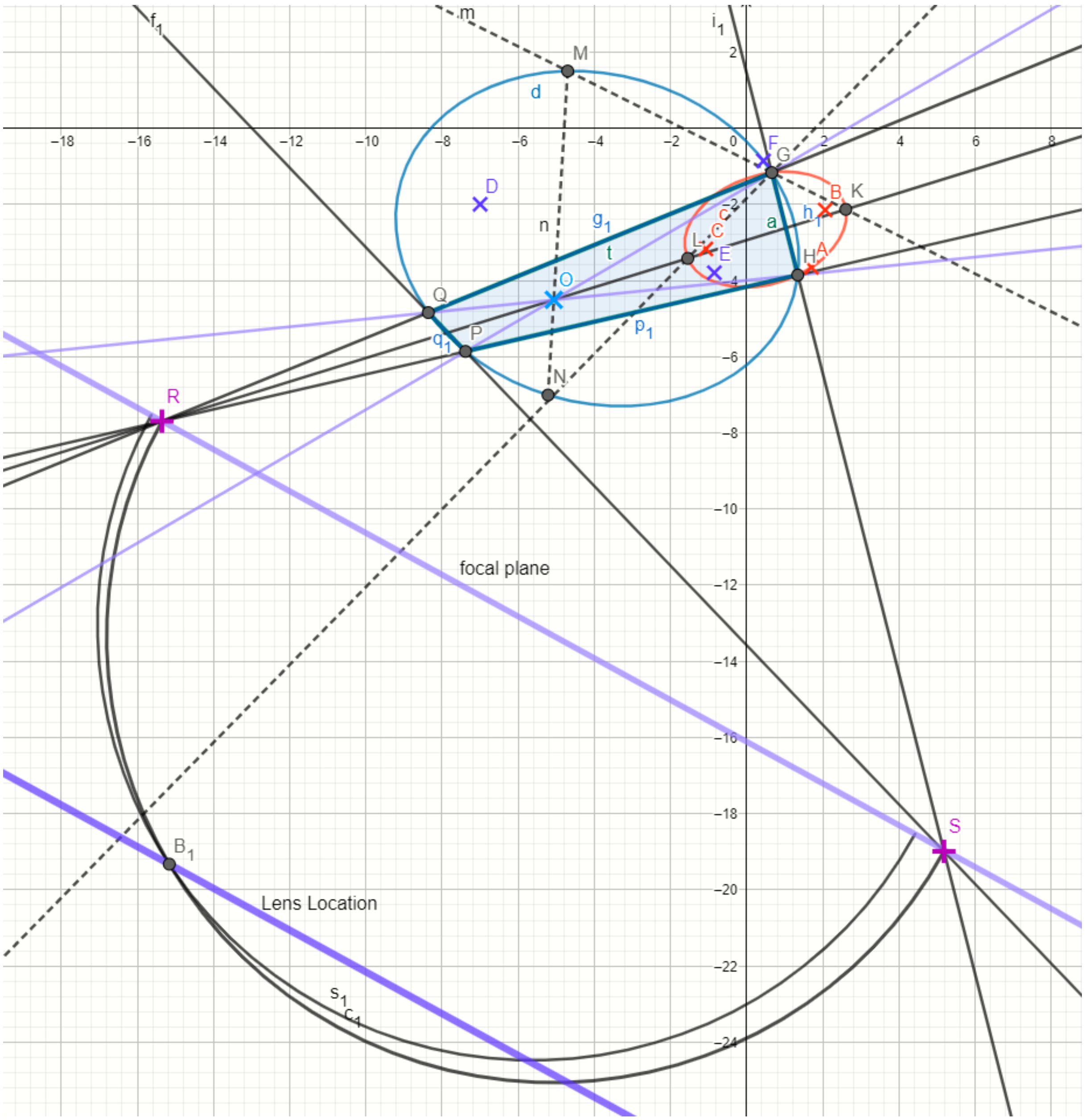
Now that we have constructed the original center of the circle O, the rest of the construction is slightly similar to 2017 Physics Cup Problem 4. Consider two arbitrary rays (lines) that pass through the center of the ellipse. Before refracting, they form an inscribed rectangle $GHPQ$ within the circle which then gets transformed into a new quadrilateral $G'H'P'Q'$ upon refraction. Let GQ and PH intersect at R and let QP and GH intersect at S. R and S must be on the focal plane as parallel lines intersect there. This then allows us to construct the focal plane of the lens. Also note that the center of the lens will lie on some point of circle ω_1 with diameter RS.





Constructing the Lens

Now that we have constructed the focal plane, we create another semicircle ω_2 by selected some more arbitrary points on the ellipse. The intersection of ω_1 and ω_2 will then be a point on the lens. The lens is, of course, parallel to the focal line so that is all we need to do. The details of the construction of the second semicircle are not presented in the following picture as it is redundant but the final construction is now laid.


focal plane
Lens Location